Problem 1

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Problem 1 *True of False: If f is continuous on the closed interval $[a, b]$, then*

$$\frac{d}{dx}\left(\int_a^b f(t)\,dt\right) = f(x)$$

Explanation. False. $\dfrac{d}{dx}\left(\displaystyle\int_a^b f(t)\,dt\right) = 0$ since $\displaystyle\int_a^b f(t)\,dt$ is a constant and $\dfrac{d}{dx}(\text{constant}) = 0$.